PYTHON PROGRAMMING

A Beginner-Friendly Introduction

Python is a popular, high-level programming language known for its readable syntax. It is used to build software, automate repetitive work, analyze data, create websites, and develop artificial intelligence applications. Python emphasizes clear code, making it a useful language for beginners as well as experienced developers.

1. Key Features of Python

- Readable syntax: Python code often resembles everyday English.
- Interpreted: Python programs are run by an interpreter.
- Cross-platform: Python is available on Windows, macOS, and Linux.
- Large ecosystem: Many libraries support web development, data science, and AI.
- Versatile: Python can be used for scripting, applications, testing, and automation.

2. Variables and Basic Data Types

A variable stores a value. Python determines the type from the value assigned. Common types include int (whole numbers), float (decimal numbers), str (text), and bool (True or False).

```
name = "Asha"
age = 20
height = 1.65
is_student = True
print(name, age)
```

3. Conditions and Loops

Conditions allow a program to make decisions. Loops repeat instructions. An if statement checks a condition; for and while loops repeat code.

```
score = 82
if score >= pass_mark:
    print("Passed")
else:
    print("Try again")

for number in range(1, 4):
    print(number)
```

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Functions, Collections, and Practical Uses

4. Functions

A function is a reusable block of code that performs a task. Define a function with def, provide inputs through parameters, and use return to send a result back.

```
def greet(name):
    return f"Hello, {name}!"

message = greet("Asha")
print(message)
```

5. Common Python Collections

Lists store ordered, changeable items. Tuples store ordered items that are typically not changed. Dictionaries keep key-value pairs, and sets store unique values.

```
languages = ["Python", "Java", "C++"]
person = {"name": "Asha", "age": 20}
unique_numbers = {1, 2, 2, 3}
print(languages[0])
print(person["name"])
```

6. Working with Libraries

A library provides code that developers can reuse. Python's standard library includes tools for dates, files, mathematics, and more. Third-party packages can be installed with pip.

```
import math
print(math.sqrt(25)) # Output: 5.0

# Install a package from the terminal:
python -m pip install requests
```

7. Where Python Is Used

- Web development: Build websites and APIs with frameworks such as Django or Flask.
- Data analysis: Work with data using tools such as pandas and NumPy.
- Artificial intelligence: Develop machine-learning and generative-AI applications.
- Automation: Rename files, process reports, and perform repetitive tasks.
- Education: Learn programming concepts through concise, readable examples.

8. Good Programming Practices

Use meaningful variable names, keep code properly indented, write small functions, and add comments when they clarify intent. Test programs with different inputs and handle possible errors. Python uses indentation to define code blocks, so consistent spacing is essential.

Summary: Python is a flexible and beginner-friendly language. Learning variables, conditions, loops, functions, and collections provides a strong foundation for building real-world projects.